

Hoff Bayesian Statistics

Jesse Mu

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These are (fully reproducible!) R Markdown lecture notes for Hoff (2009), completed as part of a 1-semester independent study course. Only Chapters 1-8 are complete right now.

Each note includes summaries of chapter sections, with math and explanations modified for clarity and the occasional link to external resources. Many figures from the book are reproduced in a ggplot/tidyverse style, and some exercises at the end of each chapter are tackled (correctness not guaranteed).

For comprehensive solutions to all exercises in the book, see <https://hoff-exercise.kaorubb.org/>.

If you find an error or would like to improve the notes, please submit a PR or open an issue!

1 Chapters

- Chapter 1: Introduction and examples¹
- Chapter 2: Belief, probability, and exchangeability²
- Chapter 3: One-parameter models³
- Chapter 4: Monte Carlo approximation⁴
- Chapter 5: The Normal Model⁵
- Chapter 6: Posterior approximation with the Gibbs sampler⁶
- Chapter 7: The multivariate normal model⁷
- Chapter 8: Group comparisons and hierarchical modeling⁸
- Chapter 9: Linear regression⁹
- Chapter 10: Nonconjugate priors and Metropolis-Hastings algorithms¹⁰

2 Final Project

As a small final project, R code for the basic binary relation version of the Infinite Relational Model was implemented, as described in Kemp et al. (2006).

- Infinite Relational Model¹¹

Hoff, Peter D. 2009. *A First Course in Bayesian Statistical Methods*. Springer Texts in Statistics. Springer. <https://doi.org/10.1007/978-0-387-92407-6>.

¹1.qmd

²2.qmd

³3.qmd

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¹⁰10.qmd

¹¹irm.qmd

Kemp, Charles, Joshua B. Tenenbaum, Thomas L. Griffiths, Takeshi Yamada, and Naonori Ueda. 2006. “Learning Systems of Concepts with an Infinite Relational Model.” *Proceedings of the National Conference on Artificial Intelligence (AAAI)* 21: 381–88. <http://web.mit.edu/cocosci/archive/Papers/Kemp-et al-AAAI06.pdf>.